ModelIt!

Teacher's Guide

Why Do Marbles Crash and Bounce?

The Surprising Science of Energy in Collisions

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-PS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 4-PS3-3

Ask questions and predict outcomes about the changes in energy that occur when objects collide.

Three-Dimensional Learning

Science Practice	**Asking Questions and Defining Problems** — Students ask questions about what happens to energy during collisions and predict outcomes before testing with their model.
Disciplinary Core Idea	**PS3.B Conservation of Energy and Energy Transfer** — Energy is present whenever there are moving objects, sound, light, or heat. When objects collide, energy can be transferred from one object to another.
Crosscutting Concept	**Energy and Matter** — Students track how energy moves from one marble to another during collisions, understanding that energy is transferred, not created or destroyed.

CAST Testing Alignment

MULTIPLE CHOICE:

A fast-rolling marble hits a marble that is sitting still. After the collision, the first marble slows down and the second marble starts moving. Which best explains what happened to the energy?

CONSTRUCTED RESPONSE:

Two students each roll a marble toward a target. Student A rolls slowly and Student B rolls quickly. Explain what will be different about how each marble hits the target, using the words energy and speed in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The science of collisions is fundamental to understanding how energy works in the physical world. From billiard balls to car crashes to particle physics, collisions reveal the basic rules that govern how energy moves between objects.

Energy in Motion

Any moving object has kinetic energy — the energy of motion. The amount of kinetic energy depends on two things: the object's mass (how heavy it is) and its speed. Doubling the speed of an object QUADRUPLES its kinetic energy (energy is proportional to speed squared). This is why high-speed car crashes are so much more dangerous than low-speed ones — a car going 60 mph has four times the kinetic energy of one going 30 mph.

Types of Collisions

There are two main types of collisions. In an elastic collision (like billiard balls), the objects bounce apart and kinetic energy is conserved — the total kinetic energy before and after is the same. In an inelastic collision (like a car crash), some kinetic energy is converted to other forms — heat, sound, and deformation. Most real-world collisions are partially inelastic.

Conservation of Energy

The law of conservation of energy states that energy cannot be created or destroyed — only transformed from one type to another. In a marble collision, the kinetic energy of the moving marble transfers to the stationary marble. Some energy also converts to sound (the click you hear) and a tiny amount of heat (from friction). If you add up all the energy after, it equals the energy before.

Real-World Applications

Understanding collisions is critical for car safety engineering (crumple zones absorb collision energy), sports equipment design (helmets, golf clubs, baseball bats), and even space science (spacecraft use gravity-assist maneuvers that are essentially collisions with a planet's gravitational field). Newton's Cradle — the classic desk toy with swinging metal balls — perfectly demonstrates energy transfer through collisions.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify marble speed, collision force, and energy transfer as the key components of the collision system.

E **Establish Relationships**
Students determine that marble speed positively affects collision force (faster marble = stronger collision), and collision force positively affects energy transfer (stronger collision = more energy moves to the second marble).

V **Visualize & Model**
Students build a model showing how the speed of a rolling marble determines the force of collision and how much energy transfers to the marble it hits.

E **Evaluate Outcomes**
Students run gentle tap and fast roll scenarios to observe how speed dramatically changes the collision outcome.

R **Revise & Extend**
Students add marble size or surface friction to explore more complex collision dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Everyone has played with marbles. But have you ever wondered — what ACTUALLY happens when they crash?"*

Watch For

Roll two marbles together on a desk and let students observe. Ask: Where did the energy go?

Slide 2: Learning Objectives

2 min

Say: *"Today we are going to figure out the secret science behind every crash, bounce, and bump."*

Watch For

Have students read objectives aloud. Pre-teach collision and energy transfer.

Slide 3: Big Question

3 min

Say: *"When a moving marble hits a still marble and the still one shoots away — where did THAT energy come from?"*

Watch For

Quick demonstration with marbles. Have students describe what they see.

Slide 4: LEVER Framework

2 min

Say: *"We will build a model to understand exactly what happens to energy in a collision."*

Watch For

Preview LEVER steps. Connect to marble play experiences.

Slide 5: Activity 1: Components

7 min

Say: *"What determines how hard marbles crash and how much energy moves between them?"*

Watch For

Guide component sorting. Discuss why marble speed is the starting input.

Slide 6: Activity 2: Connections

7 min

Say: *"Faster marble means bigger crash. Bigger crash means more energy moves. Both positive!"*

Watch For

Help students draw arrows. Both relationships in this system are positive.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us crash some marbles! First predict, then test with our model."*

Watch For

Students predict, then run scenarios. Compare gentle tap vs fast roll side by side.

Slide 8: Discoveries

4 min

Say: *"Energy is never created or destroyed — it just moves from one marble to another!"*

Watch For

Lead discussion about conservation of energy. Connect to real-world examples.

Slide 9: STEM Challenge

8 min

Say: *"Now design the ultimate marble run where collisions transfer energy to hit a target!"*

Watch For

Distribute materials. Groups design and test marble run collision courses.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Marble Speed
- **INTERNAL:** Collision Force, Energy Transfer

Relationships

- **Marble Speed to Collision Force:** POSITIVE (+) — A faster marble has more kinetic energy, which produces a stronger collision force when it hits another marble. Doubling the speed actually quadruples the energy.
- **Collision Force to Energy Transfer:** POSITIVE (+) — A stronger collision transfers more energy from the moving marble to the stationary marble. The harder the crash, the faster the second marble shoots away.

Key Discoveries

- Faster-moving marbles have more energy and create stronger collisions
- When a moving marble hits a still marble, energy transfers — the moving one slows down and the still one starts moving
- Energy is never lost in a collision — it just moves from one object to another (and some becomes sound and heat)
- The speed of the second marble after the collision depends on how much energy was transferred

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Energy disappears when things stop moving"	Energy NEVER disappears. When a marble stops, its kinetic energy has been transferred to other marbles, converted to sound (the click), converted to heat (from friction), or absorbed by the surface it is rolling on. If you add up ALL the forms of energy after a collision, the total is exactly the same as before. This is one of the most important laws in all of science — the conservation of energy.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier objects always win in collisions"	While heavier objects have more momentum at the same speed, the outcome of a collision depends on BOTH mass and speed. A small, fast marble can transfer more energy than a large, slow marble because kinetic energy increases with the SQUARE of speed. Doubling speed quadruples the energy, but doubling mass only doubles the energy. Speed matters more!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Collisions only happen between solid objects"	Collisions happen between ALL types of matter — including liquids and gases. When wind hits a sail, that is a collision between air molecules and fabric. When a raindrop hits a puddle, that is a liquid-liquid collision. Even light can collide with matter (that is how solar panels work). The science of energy transfer in collisions applies everywhere, not just to marbles and balls.